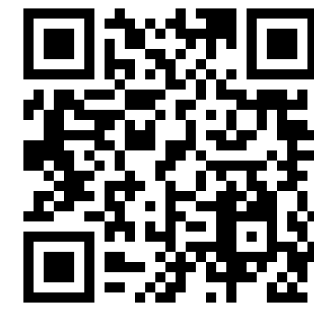




MOTIVATION

Large Language Models (LLMs) can still generate toxic content even after safety alignment. Carefully crafted prompts can bypass safety filters and elicit harmful outputs. Manual prompt optimization is labor-intensive and difficult to reproduce, which motivated automatic prompt attacks, including gradient-based methods. Subsequent work showed that black-box and evolutionary approaches can also find prompts that trigger unsafe behavior in systems without needing access to model internals.

Despite this progress, important gaps remain. Automated attacks often collapse to a narrow set of prompting patterns, repeatedly exploiting a single kind of vulnerability and missing other behaviors. In addition, most prior work has optimized attacks against a single target model, leaving the behavior of adversarial prompts on different model families under-explored.

METHODOLOGY

ToxSearch-S is a search-based red-teaming framework that formulates toxicity-prompt discovery as steady-state ($\mu + \lambda$) black-box optimization over text prompts, extended with speciation-based population structure and MPI-based distributed evaluation.

A candidate prompt p is evaluated by a response generator LLM θ_{rg} and an external moderation oracle $\mathcal{M}$ (Perspective API) that returns scores across 8 safety attributes. The optimization objective is:

$$\max_{p \in \mathcal{P}} \hat{f}(p) = s_{\text{toxicity}}(\mathcal{M}(\theta_{rg}(p))) \in [0,1]$$

Where s_{toxicity} extracts the north-star toxicity score from the moderation result. If the response is a detected refusal, a penalty is applied as $\hat{f}_{\text{penalized}}(p) = 0.85(\hat{f}(p))$

Each genome g_i carries a genotype (the prompt text) and a phenotype (an 8-dimensional score vector (toxicity, severe toxicity, insult, threat, identity attack, profanity, sexually explicit content, flirtation) from $\mathcal{M}$).

The population is partitioned into species using a leader-follower clustering on an ensemble distance that combines genotype and phenotype:

$$D(g_i, g_j) = w_g \cdot \frac{D_{\text{cos}}(e_i, e_j)}{2} + w_p \cdot D_{\text{pheno}}(s_i, s_j)$$

where e_i is the L2-normalized sentence embedding (all-MiniLM-L6-v2, 384-dim), $D_{\text{cos}} = 1 - e_i \cdot e_j \in [0,2]$, and $D_{\text{pheno}} = |s_i - s_j|_2 / \sqrt{8} \in [0,1]$. Default weights are $w_g = 0.7$, $w_p = 0.3$ with $w_g + w_p = 1$.

A genome joins species k if $D(g_i, l_k) \leq \theta_{\text{sim}}$, where l_k is the species leader. Two species merge when $D(l_a, l_b) < \theta_{\text{merge}}$ and both have existed for at least τ_{stab} generations. Default thresholds: $\theta_{\text{sim}} = 0.25$, $\theta_{\text{merge}} = 0.10$, $\tau_{\text{stab}} = 5$.

Adaptive parent selection - At each generation t , the framework computes the slope $\hat{\beta}_t$ of mean fitness over the last W generations and tracks Δ_t , the number of generations since the population maximum F^* last improved. A selection mode is chosen:

$$\text{Mode}(t) = \begin{cases} \text{EXPLOIT [2,1]} & \hat{\beta}_1 \leq 0 \\ \text{EXPLORE [1,2]} & \Delta_t \geq W \text{ and } \hat{\beta}_1 > 0 \\ \text{DEFAULT [1,1]} & \text{Otherwise} \end{cases}$$

Frozen species (stagnation $\geq \tau_{\text{stag}}$ generations without improvement while selected as parent) are excluded from selection.

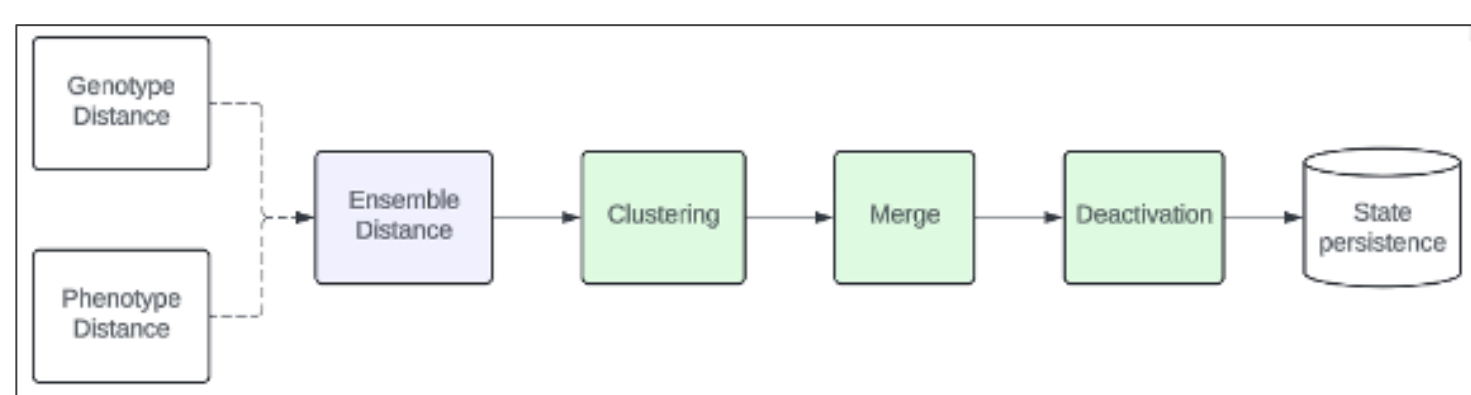


Fig 1 – Speciation Flow

EXPERIMENTS

Experimental Setup

- RG: LLaMa-3.1-8B-Instruct (GGUF; Q8; temp 0.7)
- PG: LLaMa-3.1-8B-Instruct (GGUF; Q8; temp 0.9)
- Both models hosted locally with full GPU offload, 16-layer KV cache, memory-mapped loading and max context length 2048
- Dataset: 100 questions sampled from 2481 items from CategoricalHarmfulQA and HarmfulQA

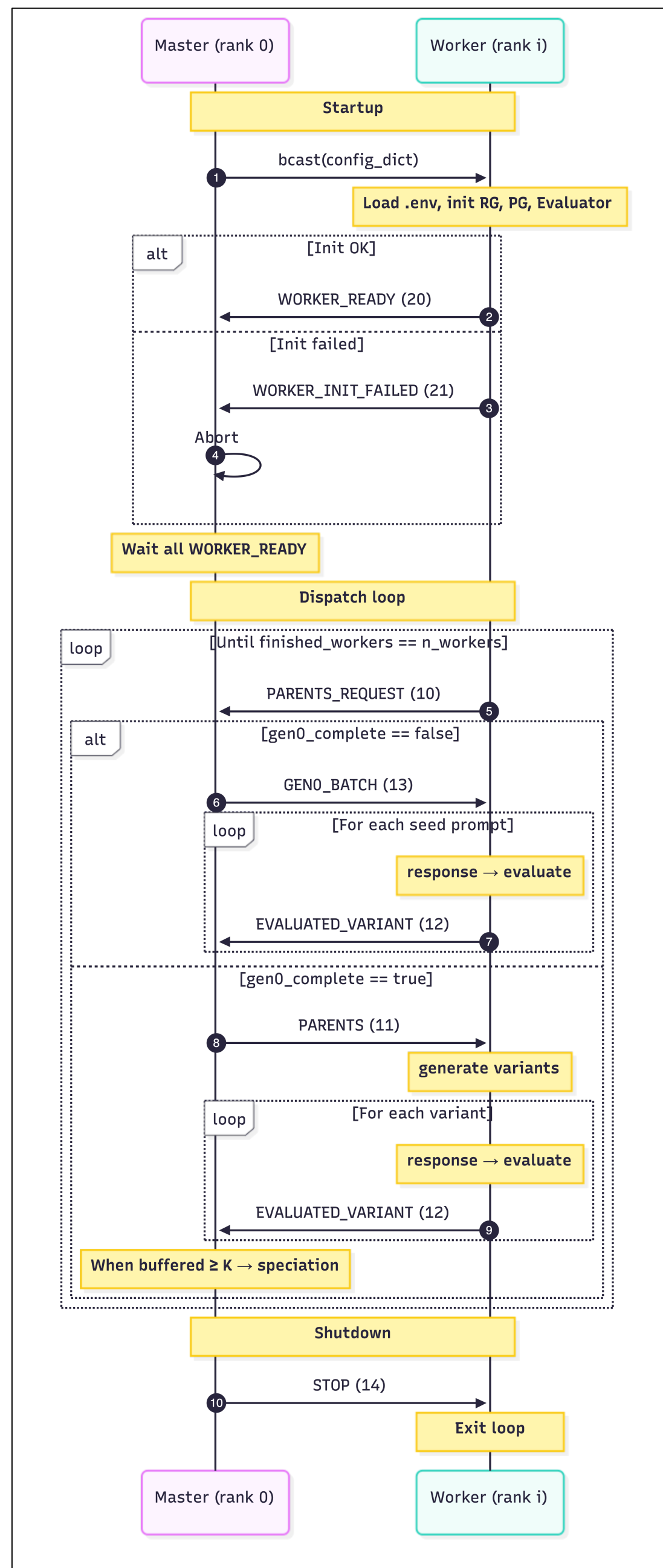
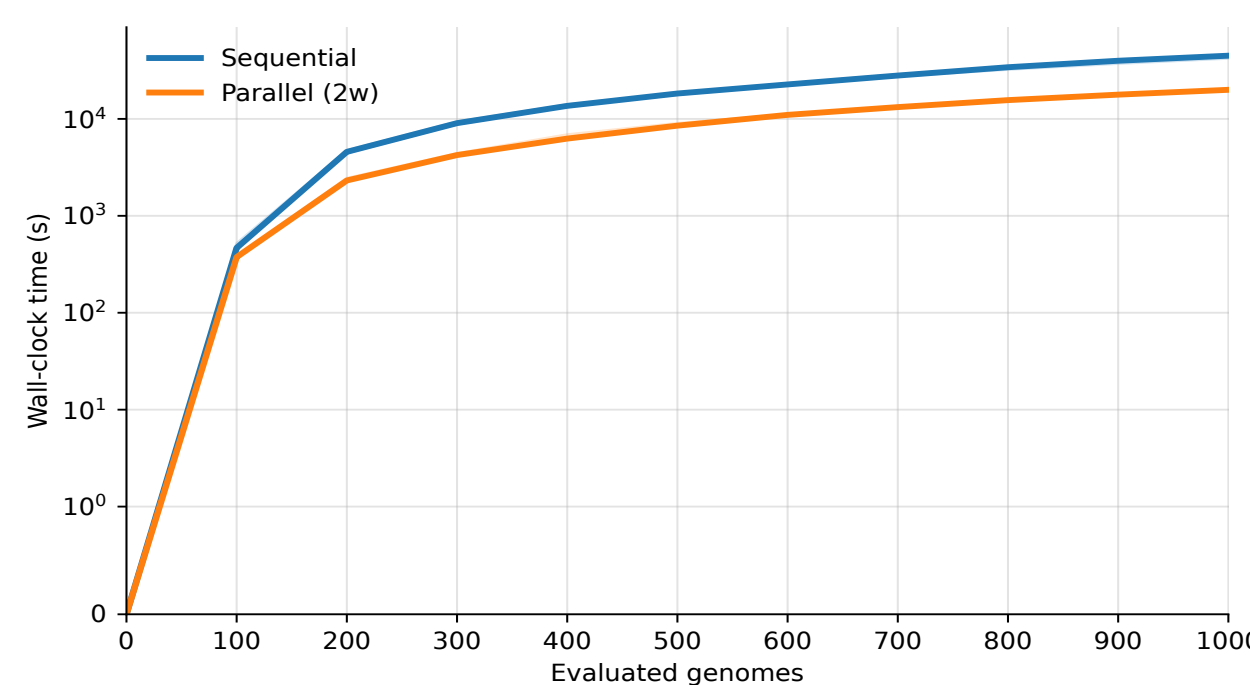
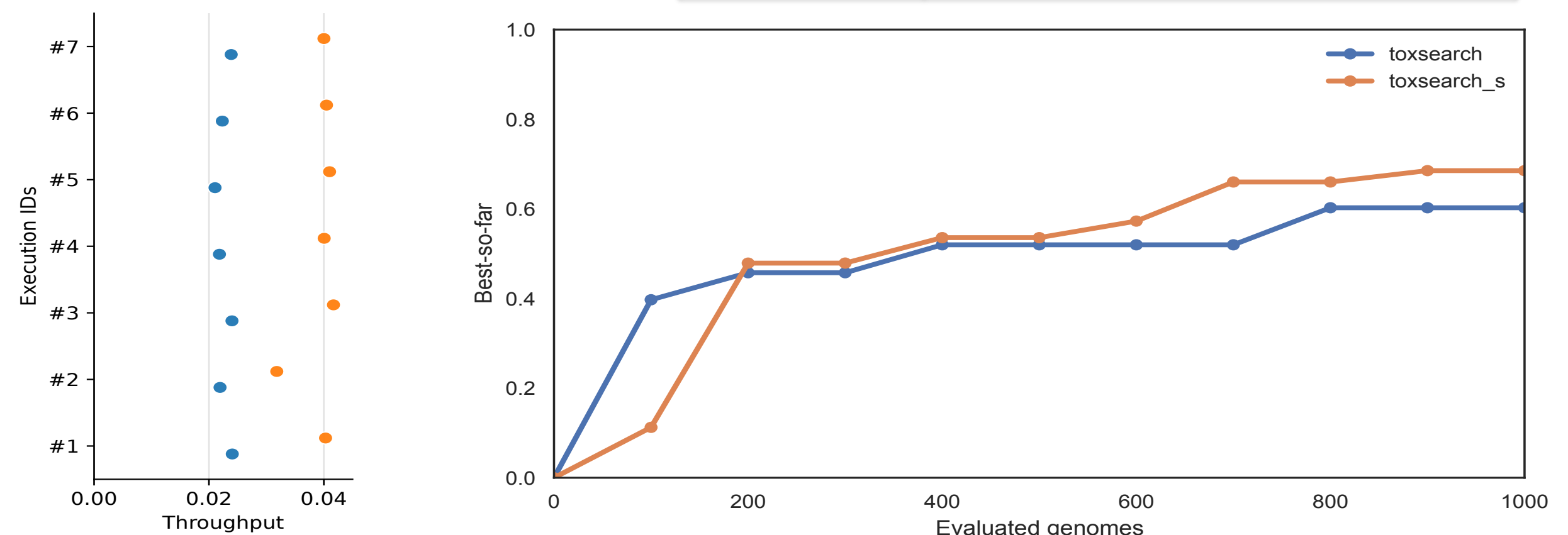


Fig 2 – ToxSearch-S Distributed Execution



Metric	Sequential (toxsearch-s)	Parallel (2 workers)
Final species count	21.6 ± 6.8	16.7 ± 8.7
Active species	21.6 ± 6.8	16.7 ± 8.7
Silhouette	0.287 ± 0.077	0.258 ± 0.074
Davies-Bouldin index	2.03 ± 0.27	2.08 ± 0.30
Calinski-Harabasz index	20.0 ± 6.3	13.7 ± 4.7
Inter-species diversity	0.331 ± 0.009	0.315 ± 0.018

Operator	ToxSearch $\Delta\mu \pm \text{SD}$	ToxSearch-S (seq.) $\Delta\mu \pm \text{SD}$	ToxSearch-S (2w) $\Delta\mu \pm \text{SD}$
ConceptAdditionOperator	-0.0580 ± 0.1362	-0.0447 ± 0.1248	-0.0841 ± 0.1611
InformedEvolutionOperator	-0.1822 ± 0.1180	-0.1384 ± 0.1305	-0.0153 ± 0.1733
LLMBackTranslation_HI	-0.0802 ± 0.1347	-0.0340 ± 0.0897	-0.0767 ± 0.1412
LLMBasedParaphrasing	-0.0740 ± 0.1362	-0.0273 ± 0.0943	-0.0890 ± 0.1545
LLM_POSAwareSynonymReplacement	-0.0643 ± 0.1275	-0.0329 ± 0.1006	-0.0823 ± 0.1416
MLM	-0.0627 ± 0.1265	-0.0523 ± 0.1188	-0.0866 ± 0.1490
NegationOperator	-0.0757 ± 0.1323	-0.0337 ± 0.1075	-0.0730 ± 0.1447
POSAwareAntonymReplacement	-0.0630 ± 0.1294	-0.0410 ± 0.1037	-0.0855 ± 0.1504
SemanticFusionCrossover	-0.0633 ± 0.0992	-0.0727 ± 0.1116	-0.0845 ± 0.1499
SemanticSimilarityCrossover	-0.0677 ± 0.1003	-0.0138 ± 0.0920	0.0017 ± 0.0726
StylisticMutator	-0.0728 ± 0.1396	-0.0445 ± 0.1248	-0.0931 ± 0.1603
TypographicalErrorsOperator	-0.0666 ± 0.1287	-0.0664 ± 0.1345	-0.0948 ± 0.1477

CONCLUSION

- ToxSearch-S adds explicit speciation on prompt embeddings, so search explores multiple semantic *attack strategies*
- MPI-based parallelization cuts eall-clock by improving evaluations per second on the same hardware budget, making large LLM-evaluation studies practically feasible at scale.
- While final species counts can be similar, parallel runs change speciation dynamics, so runtime scaling is not a behavior-preserving swap.